EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 2

PRINTABLE PRACTICE WORKSHEET

EDUSAMBAM, working on building your Foundational Concepts. Visit edusambam.com for Auto-graded Self-assessment Experience.

YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP

PHYSICAL

1. Download any practice test →
2. Solve within assigned time →
3. Submit it to your Teacher/Examiner →
4. Get assessed and improve

ONLINE

1. Visit edusambam.com →
2. Open Assessment Arena →
3. Solve online papers →
4. Get auto-assessed and improve

Candidate Name:		**Roll No.:**	
Class / Grade:		**Date:**	
School / Institution:		**Test Centre:**	

Subject	Biology	**Topic**	Biology Comprehensive Practice Test 2	**Total Marks**	100
Total Questions	45	**Time Allowed**	1 Hour (60 minutes)	**Version**	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	**45**	**—**	**100**

Section A — Multiple Choice — 30 marks

1. The first line of defence against many pathogens includes: **Basic · 2 pts**

A. Antibodies made after infection
B. Skin and mucous barriers
C. Memory B cells made by earlier exposure
D. Red blood cells carrying oxygen

Answer: ☐A ☐B ☐C ☐D

2. Innate immunity is best described as: **Basic · 2 pts**

A. A response that occurs only after a vaccine has been given to the person
B. A defence based entirely on antibodies made by plasma cells
C. A system that contains no cells, only chemicals in the blood and tissues
D. A rapid, broadly acting defence present before a specific infection

Answer: ☐A ☐B ☐C ☐D

3. Antibodies are produced by: **Basic · 2 pts**

A. Activated B-lineage cells, especially plasma cells
B. Red blood cells made in the bone marrow
C. Platelets that help blood to clot
D. Alveolar cells that line the air sacs of the lungs

Answer: ☐A ☐B ☐C ☐D

4. A key feature of adaptive immunity is: **Basic · 2 pts**

A. Absence of receptors on immune cells
B. Immediate action without any recognition
C. Specific recognition and immunological memory
D. Inability to respond to the same pathogen again

Answer: ☐A ☐B ☐C ☐D

5. Vaccination works primarily by: **Basic · 2 pts**

A. Destroying every microorganism that is already inside the body
B. Training the immune system to recognise a target without causing the disease
C. Replacing the body's white blood cells with new ones
D. Making all pathogens permanently unable to mutate

Answer: ☐A ☐B ☐C ☐D

6. A booster dose can be useful because: **Intermediate · 2 pts**

A. The first vaccine permanently disappears from the body after a few hours
B. It changes all pathogens into harmless organisms
C. It removes all immune cells so that new ones can form
D. Immune memory can be strengthened and antibody responses can rise again

Answer: ☐A ☐B ☐C ☐D

7. A person infected with a virus is not necessarily helped by an antibiotic because: **Intermediate · 2 pts**

A. Antibiotics target bacterial features, not viral replication
B. Viruses are always larger than bacteria and cannot be reached
C. Antibiotics only work on the cells of plants and fungi
D. Viruses have cell walls that antibiotics cannot reach

Answer: ☐A ☐B ☐C ☐D

8. Antimicrobial resistance is strongly promoted when: **Intermediate · 2 pts**

A. All microbes in a population become resistant at the same moment
B. Antibiotics deliberately create every mutation that a microbe needs
C. Resistant variants survive and reproduce while susceptible microbes are killed
D. Resistance can never be inherited by the next generation of microbes

Answer: ☐A ☐B ☐C ☐D

9. Autoimmune disease involves: **Intermediate · 2 pts**

A. Complete absence of all immune cells in the blood
B. Immune responses directed against the body's own tissues
C. An infection that is caused only by bacteria
D. Allergic reactions to harmless foreign substances only

Answer: ☐A ☐B ☐C ☐D

10. An allergy is best understood as: **Intermediate · 2 pts**

A. A disease that is always caused by bacteria
B. A complete lack of antibodies in every person who has it
C. A genetic mutation that is present in every immune cell
D. An excessive immune response to a usually harmless substance

Answer: ☐A ☐B ☐C ☐D

11. Which immune cell type can directly kill infected or abnormal target cells? **Advanced · 2 pts**

A. Red blood cells
B. Platelets
C. Cytotoxic T cells
D. Osteocytes

Answer: ☐A ☐B ☐C ☐D

12. Inflammation can be useful because it: **Advanced · 2 pts**

A. Helps recruit immune cells and boosts local defence
B. Always shows that an autoimmune disease is present in the body
C. Prevents any immune response from starting at the site
D. Occurs only in the tissues of the brain

Answer: ☐A ☐B ☐C ☐D

13. Why can a second exposure to the same antigen sometimes produce a faster response? **Advanced · 2 pts**

A. Pathogens always become weaker after the first exposure to them
B. The skin becomes permanently thicker after every infection
C. Red blood cells remember the antigens that they have met
D. Memory cells formed during the first response can respond rapidly

Answer: ☐A ☐B ☐C ☐D

14. A vaccine with high effectiveness can reduce disease spread because: **Advanced · 2 pts**

A. Vaccinated people can never carry any disease-causing microorganism
B. Fewer susceptible people can mean fewer chances for the pathogen to spread
C. It removes every pathogen from the environment around us
D. It guarantees lifelong sterilising immunity in every person

Answer: ☐A ☐B ☐C ☐D

15. Which statement best integrates immunity and public health? **Advanced · 2 pts**

A. Public health has no biological component and depends only on rules
B. Only antibody levels in one person matter for disease control
C. Individual protection and population-level spread can influence each other
D. Vaccination changes only the genes of the vaccinated person

Answer: ☐A ☐B ☐C ☐D

Section B — True / False **20 marks**

16. The immune system can distinguish many molecular patterns using specialised receptors. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

17. Innate immunity is completely absent until after vaccination. **Basic · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

18. Antibodies are proteins that can bind specific targets. **Basic · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

19. All inflammation is harmful. **Basic · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

20. Vaccines are designed to induce immune memory against particular targets. **Basic · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

21. Antibiotics are effective treatments for every viral infection. **Intermediate · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

22. Antimicrobial resistance can spread through microbial populations. **Intermediate · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

23. Autoimmune reactions can involve immune recognition of self molecules as targets. **Advanced · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

24. A person with immune memory can never become infected again. **Advanced · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

25. Herd or community protection can depend on how many people are susceptible and how efficiently an infection spreads. **Advanced · 2 pts**
A. True **B.** False
Answer: ☐ **A** ☐ **B**

Section C — Fill in the Blank — 20 marks

26. The first line of physical defence includes the skin and _____ membranes. **Intermediate · 2 pts**
Answer: _____

27. The branch of immunity that includes rapid, broad defences is _____ immunity. **Intermediate · 2 pts**
Answer: _____

28. Cells that secrete large amounts of antibodies are called _____ cells. **Intermediate · 2 pts**
Answer: _____

29. Long-lived cells that contribute to faster secondary responses are _____ cells. **Intermediate · 2 pts**
Answer: _____

30. A substance or molecular structure recognised by specific immune receptors is an _____. **Intermediate · 2 pts**
Answer: _____

31. A preparation that trains the immune system to recognise a pathogen without causing the disease is called a _____. **Advanced · 2 pts**
Answer: _____

32. The inappropriate immune response to harmless substances seen in hay fever is an _____. **Advanced · 2 pts**
Answer: _____

33. Immune attack against the body's own tissues is characteristic of _____ disease. **Advanced · 2 pts**
Answer: _____

34. The ability of microbes to survive drugs that once killed them is called antimicrobial _____. **Advanced · 2 pts**
Answer: _____

35. The movement of an infection through a population depends partly on transmission and the number of _____ people. **Advanced · 2 pts**

Answer: _____

Section D — Column Matching — 10 marks

36–40. Match each term in Column A with the correct description in Column B. **Advanced · 10 pts**

Column A	Answer	Column B
Antibody	____	**A.** Exposure designed to train immune memory
Memory cell	____	**B.** Reduced susceptibility of a microbe to a drug
Vaccine	____	**C.** Specific protein that binds an antigen
Autoimmunity	____	**D.** Supports a faster secondary immune response
Antimicrobial resistance	____	**E.** Immune attack against self

Section E — Short Answer — 20 marks

41. Compare innate and adaptive immunity in terms of speed, specificity and memory. **Advanced · 4 pts**

42. Explain why vaccination can protect an individual without giving them the disease the vaccine is designed to prevent. **Advanced · 4 pts**

43. Explain how antimicrobial resistance can increase in a bacterial population after antibiotic exposure. **Advanced · 4 pts**

44. Why does autoimmune disease differ conceptually from infection? **Advanced · 4 pts**

45. Explain how individual vaccination and population transmission can interact. **Advanced · 4 pts**

Candidate Signature: _____ **Invigilator Signature:** _____

FOR OFFICE USE ONLY	
Marks Obtained: / 100	Checked By:
Result / Grade:	Date Checked:

EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 2 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

1. B
2. D
3. A
4. C
5. B
6. D
7. A
8. C
9. B
10. D
11. C
12. A
13. D
14. B
15. C

Section B — True / False

16. True
17. False
18. True
19. False
20. True
21. False
22. True
23. True
24. False
25. True

Section C — Fill in the Blank (accepted answers)

26. mucous / mucus
27. innate
28. plasma / plasma b
29. memory
30. antigen / antigens
31. vaccine / vaccines
32. allergy / allergies / allergic reaction
33. autoimmune / auto-immune / auto immune
34. resistance
35. susceptible / susceptibles

Section D — Column Matching

36. Antibody — C
37. Memory cell — D
38. Vaccine — A
39. Autoimmunity — E
40. Antimicrobial resistance — B

Section E — Short Answer (Model Answers)

41. Innate immunity responds within minutes to hours, is broad and non-specific, and has no lasting memory. Adaptive immunity takes days to develop at first, is specific to particular antigens, and forms memory cells that give a faster, stronger response on later exposure.
Marking: 1 mark: innate is faster to respond, adaptive slower at first; 1 mark: innate is broad or non-specific; 1 mark: adaptive is specific to particular antigens; 1 mark: adaptive immunity forms memory (innate does not).

42. A vaccine contains a weakened or inactivated pathogen, a part of it, or instructions to make an antigen, so it cannot cause the disease. The immune system recognises the antigen, responds and forms memory cells, so a later infection by the real pathogen meets a faster, stronger response.
Marking: 1 mark: vaccine contains a weakened/inactivated pathogen, a part of it, or its antigen; 1 mark: the immune system recognises it and responds; 1 mark: memory cells are formed for later protection; 1 mark: the vaccine cannot cause the disease itself.

43. Within a bacterial population some cells already carry mutations that make them resistant. An antibiotic kills the susceptible bacteria, but the resistant ones survive and reproduce, passing resistance to their offspring (and sometimes to other bacteria). Over time the population contains more resistant bacteria.
Marking: 1 mark: variation: some bacteria are already resistant (for example by mutation); 1 mark: the antibiotic kills susceptible bacteria; 1 mark: resistant bacteria survive and reproduce; 1 mark: resistance is passed on, so it becomes more common.

44. In an infection, an outside pathogen such as a bacterium or virus invades and the immune system responds to remove it. In autoimmune disease no pathogen is needed: the immune system wrongly attacks the body's own healthy tissues, so the immune response itself causes the damage. It is not caught from other people.
Marking: 1 mark: autoimmune: the immune system attacks the body's own tissues; 1 mark: infection: caused by an outside pathogen; 1 mark: autoimmune disease is a misdirected response, not caused by a pathogen or caught from others; 1 mark: the immune response itself causes the damage.

45. Vaccination makes individuals less susceptible to infection, so fewer people can catch and pass on the pathogen. This reduces its spread through the population and, when enough people are protected, can indirectly protect unvaccinated or vulnerable people (herd or community protection).
Marking: 1 mark: vaccination protects the individual; 1 mark: fewer susceptible people; 1 mark: the pathogen spreads less, so transmission chains are broken; 1 mark: community (herd) protection can help unvaccinated or vulnerable people.